When can a conditional generative model be trusted as a posterior?

Identifiability, bandwidth, and out-of-fold recalibration of flow-matching ensembles

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Abstract

Samples from conditional diffusion and flow-matching models are increasingly read as draws from a posterior: ensembles of weather, reconstructions in inverse problems, analyses in data assimilation. Some of these ensembles are well calibrated (GenCast) and many are not. We give a simple explanation. A conditional flow-matching model trained on pairs (x_i, y_i) with one realisation per condition is a nonparametric conditional-density estimator with an implicit bandwidth over the condition: it reproduces $p(y \mid x)$ only if the training conditions are dense at the scale over which $p(y \mid x)$ varies. If the condition identifies its single training realisation, the empirical minimiser of *any* strictly proper objective is a point mass (Theorem 1); the model's spread is a bandwidth artefact that shrinks with training. We prove that noise injected into the condition and resampled at every step turns the population minimiser into the exact Bayesian posterior under the empirical prior, i.e. a Nadaraya–Watson estimator with bandwidth σ (Theorem 2), which yields a sharp criterion—calibration iff σ exceeds the nearest-neighbour distance of the conditions—and its price in information. We further show that the collapse degrades the marginal scale but not the dependence structure: the copula of a mixture of real realisations converges to the true copula (Theorem 3). This justifies an out-of-fold recalibration that replaces the marginals cell by cell and keeps the generator's copula, with conformal-type guarantees. Controlled experiments with known posteriors confirm every statement. In an operational sea-ice assimilation problem with real satellite-track geometry, whose conditions provably have no neighbours (six independent years), the recalibrated ensemble is calibrated on the development year (CRPS -18%, spread/skill 0.99, flat rank histograms, correct atoms) and transfers unchanged to a sealed test year of 200 consecutive days (CRPS -44%, spread/skill 0.98, 189/200 days improved), while removing the identifying part of the condition yields calibration by construction at a cost in skill.

1 Introduction

Conditional generative models (score-based diffusion, flow matching) are trained to transport noise into $p(y \mid x)$ and, in the population limit, they do [Lipman et al., 2023]. In practice they are trained on finite data in which each condition x_i comes with a single realisation y_i: one future for each initial state, one image for each degraded measurement, one analysis for each background. The ensemble obtained by sampling the trained model several times is then read as a posterior. This reading is correct for GenCast [Price et al., 2025] and incorrect for many other systems, including the sea-ice assimilation model studied here, whose analysis ensemble was under-dispersed by a constant factor in every regime and whose spread *decreased* monotonically with training.

The explanation we propose is elementary and, we think, useful. A trained network with finite capacity acts on the condition as a smoother: its output law at a new x is a weighted mixture of the training realisations of *neighbouring* conditions. When neighbours exist at a scale where $p(y \mid x)$ does not change—58 000 initial states with a smooth chaotic response, as in GenCast—the

mixture is the right distribution. When the condition identifies its realisation—a background field that occurs once, a measurement that fingerprints its image—no neighbours exist, the empirical minimiser is a point mass, and the spread one observes is the width of the network's smoothing window, which closes as training proceeds. No change of objective helps, because every strictly proper objective has the same minimiser.

Contributions. (i) We formalise the smoother view: Theorem 1 (point-mass minimiser), Theorem 2 (condition noise resampled per step = exact posterior under the empirical prior, i.e. an explicit bandwidth), Theorem 3 (the copula of the mixture survives the collapse of its marginals). (ii) Controlled experiments with analytically known posteriors confirm the three statements and give a quantitative criterion: calibration when the noise bandwidth exceeds the nearest-neighbour distance of the training conditions, or when the conditions are dense; the price is information (Fig. 1). (iii) An out-of-fold recalibration layer that combines classical blocks (kriging of observation innovations, out-of-fold cell quantiles, ensemble-copula coupling) and is calibrated with respect to the cell σ-algebra by construction; the generator supplies the conditional mean and the copula. (iv) A real case with a diagnosis: sea-ice data assimilation with last-year backgrounds and real altimeter-track geometry. Its conditions have no neighbours (nearest training background at 0.47 of the anomaly scale, median), the pure model collapses, and the recalibrated ensemble is calibrated out of sample; a model trained without the identifying part of the condition is calibrated by construction at a cost in skill.

2 Setting and the smoother view

Let $x \in \mathcal{X}$ be a condition and $y \in \mathbb{R}^m$ a target with conditional law $p(y \mid x)$. Conditional flow matching (CFM) learns a velocity field $v_\theta(y_t, x, t)$ by minimising

$$\mathcal{L}(\theta) = \mathbb{E}_{i,\,t,\,y_0}\big\|v_\theta(y_t, x_i, t) - (y_i - y_0)\big\|^2, \qquad y_t = (1-t)y_0 + t\,y_i,\ y_0 \sim \mathcal{N}(0, I), \tag{1}$$

and samples by integrating $\dot{y} = v_\theta(y, x, t)$ from y_0. In the population limit over the joint law $p(x, y)$ the minimiser of (1) is the marginal velocity field whose flow transports $\mathcal{N}(0, I)$ to $p(\cdot \mid x)$ [Lipman et al., 2023, Tong et al., 2023]. Nothing in this statement concerns the finite-sample behaviour along the condition axis, and this is where calibration is decided.

Theorem 1 (point mass)**.** *Let S be a strictly proper scoring rule and let the condition x_i occur exactly once in the training set with target y_i. If the model class is unrestricted at x_i, the unique minimiser of the empirical risk at x_i is $Q(\cdot \mid x_i) = \delta_{y_i}$. For CFM the minimising velocity at x_i is the field of the single pair (y_0, y_i), whose flow transports $\mathcal{N}(0, I)$ to δ_{y_i}.*

Proof. The contribution of i to the empirical risk is $S(Q(\cdot \mid x_i), y_i) = \mathbb{E}_{Y \sim \delta_{y_i}} S(Q(\cdot \mid x_i), Y)$, minimised uniquely at $Q = \delta_{y_i}$ by strict properness. For CFM, with y_i fixed, $v^*(y_t, x_i, t) = \mathbb{E}[y_i - y_0 \mid y_t, t]$ is the marginal field of the path with $p_1 = \delta_{y_i}$ [Lipman et al., 2023, Thm. 1]. □

Theorem 1 says what the network is pulled towards at every training condition. What it outputs at a *new* condition depends on its inductive bias in x. We use the kernel-smoother idealisation

$$Q_\theta(\cdot \mid x) \approx \sum_i w_h(x, x_i)\,\delta_{y_i}, \qquad \sum_i w_h(x, x_i) = 1, \tag{2}$$

with a bandwidth h that decreases as training proceeds. One fact makes (2) more than a metaphor: it holds exactly, for the population minimiser, when the bandwidth is supplied by condition noise

(Theorem 2). For a network without explicit noise, (2) is a hypothesis; kernel descriptions of training in the lazy regime [Jacot et al., 2018] concern the velocity field, not the law of the ODE solution, and we do not use them as a proof. For the location-scale family $y = \mu(x) + s(x)z$ one obtains from (2)

$$\mathrm{Cov}[Q_\theta(\cdot \mid x)] = \sum_i w_i \, s(x_i)^2 \Sigma_0 + \sum_i w_i (\mu(x_i) - \bar{\mu})(\mu(x_i) - \bar{\mu})^\top, \tag{3}$$

which tends to the true $s(x)^2\Sigma_0$ when neighbours are dense ($N_h \to \infty$, $h \to 0$) and to 0 when $N_h := \sum_i w_i / \max_i w_i \to 1$: the model's spread is the spread of the neighbours' truths, i.e. a property of the training set's density, not of $p(y \mid x)$.

Theorem 2 (condition noise is an explicit bandwidth)**.** *Let training pairs be generated as $i \sim \mathrm{Unif}\{1, \dots, N\}$, $\varepsilon \sim \mathcal{N}(0, \sigma^2 I_d)$ fresh at every step, $\tilde{x} = x_i + \varepsilon$, $y = y_i$, and let v_σ minimise the population CFM objective over all measurable fields with respect to the law P_σ of these pairs. Then the flow of $v_\sigma(\cdot, \tilde{x}, \cdot)$ transports $\mathcal{N}(0, I)$ to*

$$Q_\sigma(\cdot \mid \tilde{x}) = \sum_i w_i(\tilde{x}) \, \delta_{y_i}, \qquad w_i(\tilde{x}) = \frac{\varphi_\sigma(\tilde{x} - x_i)}{\sum_j \varphi_\sigma(\tilde{x} - x_j)}. \tag{4}$$

Proof. P_σ has density $p_\sigma(\tilde{x}) = \frac{1}{N} \sum_i \varphi_\sigma(\tilde{x} - x_i) > 0$ and conditional law $P_\sigma(y \mid \tilde{x}) = \sum_i w_i(\tilde{x}) \delta_{y_i}$ by Bayes' rule. The conditional CFM theorem applied to P_σ (it requires only that the objective be averaged over P_σ, which fresh noise guarantees) gives that the minimiser generates $P_\sigma(\cdot \mid \tilde{x})$ for p_σ-a.e. $\tilde{x}$. □

Corollary 1. *(a) Q_σ is the Nadaraya–Watson estimator of $p(y \mid x)$ with Gaussian kernel of bandwidth σ, independently of the architecture. (b) $Q_\sigma(\cdot \mid \tilde{x})$ is exactly the posterior of y given the noisy condition under the empirical prior $\pi_N = \frac{1}{N} \sum_i \delta_{(x_i, y_i)}$: the model is calibrated for the* noisy *problem by construction, with the sharpness of the noisy problem. (c) If $(x_i, y_i) \sim p$ i.i.d. and $p(y \mid x)$ is continuous in x, then $Q_\sigma(\cdot \mid x) \to p(\cdot \mid x)$ as $N \to \infty$, $\sigma \to 0$, $N\sigma^d \to \infty$. At finite N the effective number of neighbours is $N_\sigma(x) = \sum_i w_i / \max_i w_i$; for $\sigma \ll d_{\mathrm{NN}}(x)$ one has $N_\sigma \to 1$ and Theorem 1 through the back door, for $\sigma \gg d_{\mathrm{NN}}$ averaging over neighbours. Since $d_{\mathrm{NN}} \sim N^{-1/d}$, the bandwidth needed for calibration grows with the condition dimension, and with it the information destroyed. (d) Condition noise that is drawn once and* fixed *makes $\tilde{x}_i$ a deterministic function of i and Theorem 1 applies: such augmentation does not help. (e) The mixing acts only along the noised coordinates. If the condition is $x = (b, o)$ and only o is noised while b (a background field, an observation* mask*, any identifier) stays exact and unique, then $Q_\sigma(\cdot \mid b_i, \tilde{o}) = \delta_{y_i}$: no amount of observation noise removes the identification carried by the exact part. In our sea-ice arms this concerns the real track footprint of a training date as much as the background.*

Remark 1. *Theorem 2 identifies the* target *of the augmented training problem, not the law of a finite network trained for a finite time: the network adds its own implicit smoothing, which can move it closer to the true posterior than the target itself (Section 3, three-law comparison). Nor does one realisation per condition preclude estimation in parametric or smooth nonparametric classes [Tang et al., 2024]: the statement concerns the regime in which $p(y \mid x)$ is not resolved by the training conditions at the scale of their spacing.*

Theorem 3 (the copula survives the collapse)**.** *Let $p(\cdot \mid x)$ be the law of $y = \mu(x) + s(x)z$, $z \sim F$ with continuous marginals and copula C independent of x, $s \geq s_{\min}$. Let $Q(\cdot \mid x) = \sum_i w_i p(\cdot \mid x_i)$ and $\Delta := \sum_i w_i \big(\|\mu(x_i) - \mu(x)\| + |s(x_i) - s(x)|\big) / s_{\min}$. Then the copula C_Q of $Q(\cdot \mid x)$ satisfies $\|C_Q - C\|_\infty \to 0$ as $\Delta \to 0$; for Gaussian F, $\|C_Q - C\|_\infty \leq c\sqrt{\Delta}$ with an absolute constant c.*

Proof sketch. For Gaussian F, $\mathrm{KL}\big(\mathcal{N}(\mu_i, s_i^2\Sigma_0)\,\|\,\mathcal{N}(\mu, s^2\Sigma_0)\big) = O(\|\mu_i-\mu\|^2/s^2 + (s_i/s-1)^2)$, the KL of the mixture is bounded by $\sum_i w_i \mathrm{KL}_i$ by convexity, and $\mathrm{TV} \le \sqrt{\mathrm{KL}/2}$. Convergence in total variation of joint laws with continuous marginals implies uniform convergence of the copulas by the continuity of Sklar's map [Deheuvels, 1979, Sempi, 2004]. The general location-scale case follows from the continuity of F under shift and scale. □

The marginals of Q are distorted by the second term of (3); the dependence between coordinates is inherited with error $O(\sqrt{\Delta})$. Theorem 3 concerns the mixture of real realisations, i.e. the smoother (2) *before* its marginals collapse; it does not by itself say what the copula of a collapsed network is (scale contraction does not determine a copula: $\mathcal{N}(0, \varepsilon^2 R_\rho) \to \delta_0$ for every ρ). The operational claim we test is therefore weaker and falsifiable: the generator's error is mainly marginal, and its dependence is more useful than a stationary spatial model or the dependence of another date (Section 4, ablation at identical marginals). This is the licence for ensemble-copula coupling [Schefzik et al., 2013] applied to a generative ensemble whose marginals are an artefact: replace the marginals, keep the copula.

3 Controlled experiments with known posteriors

We use $x \sim \mathrm{Unif}[-1,1]^2$, $y \mid x \sim \mathcal{N}(f(x), \sigma(x)^2 K)$ in $\mathbb{R}^{16}$ with smooth random-feature f, heteroscedastic $\sigma(x)$ and an exponential correlation K, and a three-layer MLP-CFM (256 units, 20k Adam steps, Euler-32 sampling, 32 members). Everything about $p(y \mid x)$ is known, so spread is compared with the true σ, ranks with the true law, and energy scores with fresh true draws.

Three laws. On 96 held-out conditions of the noisy-observation setting we compare the true posterior P^* (grid marginalisation, converged: 81 vs 161 grid differ by 0.004 in the mean), the augmented empirical mixture $\hat{P}_{D,\sigma}$ of Theorem 2 (sampled by its weights, no training), the Euler-32 integration of the *exact* CFM field of that mixture, and the trained network. The integrator is faithful (ODE of the mixture ≈ mixture in every cell). At $\sigma = 0.3$ the mixture is itself close to P^* (spread ratio 0.97–0.99, $N_{\mathrm{eff}} = 8$–21) and the network reproduces it: the bandwidth regime of Theorem 2. At $\sigma = 0.1$ the mixture is nearly a point mass ($N_{\mathrm{eff}} = 1.4$–1.8, spread ratio 0.58, rank TV 0.56–0.63) while the network is markedly better than its own target ($N = 2048$: spread ratio 0.90, rank TV 0.10, energy score 0.68 against 0.65 for P^* and 1.13 for the mixture). The network's implicit smoothing over x adds to the explicit bandwidth; Theorem 2 is a guaranteed floor, and what lies above it is an inductive bias that has to be measured, not assumed — it is also what shrinks along the collapse curve.

Images (teacher–student). To show that nothing here is specific to physical fields we use a procedural teacher with an exactly samplable posterior: the condition is (MNIST class, rotation $\theta \in [-45°, 45°]$, scale $s \in [0.8, 1.2]$), and $p(y \mid x)$ is a random instance of the class rendered with (θ, s) plus pixel noise. A small FiLM-conditioned UNet-CFM student (15k steps, 32 members, 200 held-out conditions) gives spread/true-spread 0.38, 0.85, 0.92 for $N = 500$, 2000, 8000 conditions with one realisation each (energy score 6.85, 4.55, 4.13 against 4.1–4.2 for the true posterior): density governs calibration. Appending a unique random 8-dimensional key to the condition at $N = 2000$ collapses the student (0.85 → 0.29; rank TV 0.13 → 0.29); resampling Gaussian noise on the key at every step restores it as the noise grows (0.64 at $\sigma = 0.5$, 0.82 at $\sigma = 1.0$), exactly as Theorem 2 predicts; with eight realisations per key the key is harmless (0.95) (Fig. 2).

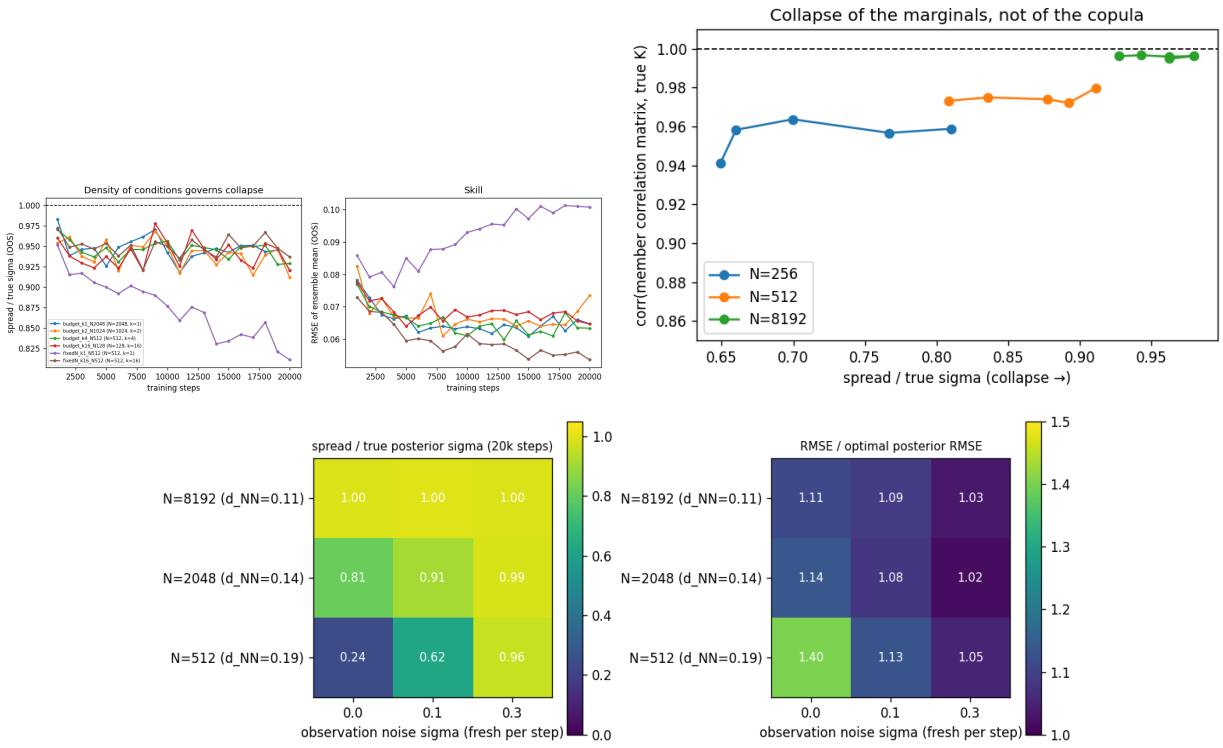


Figure 1: Controlled experiments. Top left: spread/true σ vs training steps; with a fixed budget $Nk = 2048$ the number of realisations per condition k is irrelevant (dense conditions calibrate even at $k = 1$), with $N = 512$ fixed the $k = 1$ model collapses (0.92 $\to$ 0.81, rank TV 0.06 $\to$ 0.15, out-of-sample RMSE 0.079 $\to$ 0.101) while $k = 16$ stays at 0.95. Top right: correlation between the members' correlation matrix and the true K against the collapse of the marginals: the copula is preserved (0.94–1.00). Bottom: condition = six noisily observed coordinates of y (no latent given), exact posterior computed on a grid; spread/true posterior σ and RMSE/optimal RMSE after 20k steps as a function of N (with its nearest-neighbour distance d_{NN}) and of the fresh observation noise σ. Calibration appears when $\sigma \gtrsim d_{\mathrm{NN}}$ or when conditions are dense; the optimal RMSE grows with σ.

Three further facts follow (Fig. 1). *Density, not multiplicity*: at a fixed budget the number of realisations per condition does not matter, at a fixed number of conditions the single-realisation model collapses while the 16-realisation model does not. *Noise is a bandwidth*: with the "tracks-only" condition (six noisy coordinates of y) the model collapses at $\sigma = 0$ for $N = 512$ (spread/σ 0.24, rank TV 0.44), is calibrated at $\sigma = 0.3 > d_{\mathrm{NN}} = 0.19$ (0.96, TV 0.07) and matches the exact posterior's energy score (0.90 vs 0.90), and is calibrated even at $\sigma = 0$ once $N = 8192$; the optimal RMSE rises from 0.23 to 0.37 with σ. *The copula survives*: while spread/σ falls from 0.81 to 0.65 ($N = 256$), the correlation between the members' correlation matrix and K stays at 0.94–0.96.

4 Out-of-fold recalibration

Given a generator that provides a conditional mean and a realistic copula but an artefactual spread, Theorems 1–3 suggest the following decomposition, which we implement for spatial fields with atoms

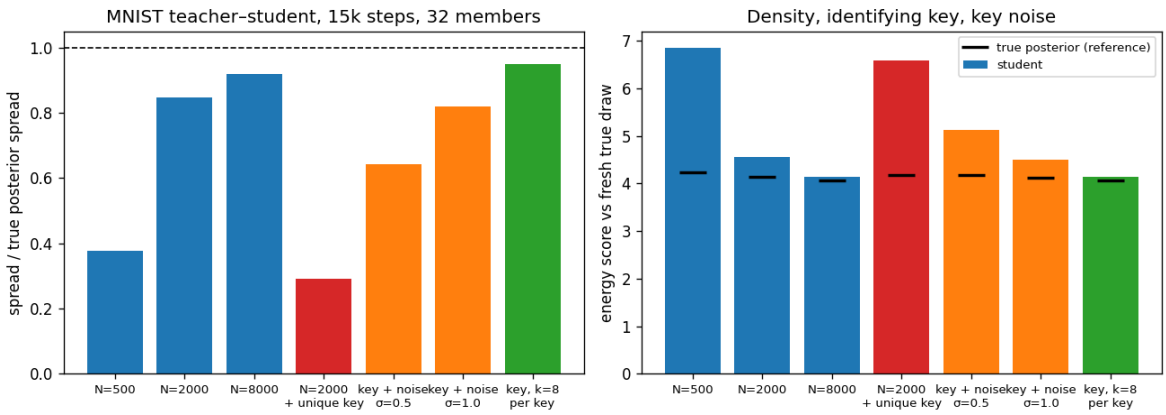


Figure 2: MNIST teacher–student with an exactly samplable posterior: spread ratio and energy score of the student for increasing numbers of conditions (blue), with a unique identifying key appended to the condition (red), with fresh Gaussian noise on the key (orange) and with eight realisations per key (green).

(exact zeros, a cap) and sparse exact observations.

1. *Latent with atoms.* Water pixels are encoded as $-\gamma\cdot$(distance to the ice edge), the cap as a censored value, so that atoms become threshold events of a continuous latent.

2. *Conditional mean.* The generator's mean latent μ is updated by kriging of the track innovations $z_{\text{obs}} - \mu$ with a two-scale stationary prior fitted out of fold and a per-date microscale nugget chosen by leave-one-out on the day's innovations; day-d observations are honoured exactly.

3. *Marginals.* The residual $r = z_{\text{truth}} - \mu_2$ is described by empirical quantiles per cell (distance to the day-d track $\times$ value of μ_2), estimated on *other* dates (5-fold, purged).

4. *Copula.* Member k receives the Gaussian score $\sqrt{1-\lambda}\,\tilde{a}_k + \sqrt{\lambda}\,g_k$, where $\tilde{a}_k$ is the generator's pixelwise-standardised anomaly pattern and g_k an independent long-scale stationary field; $u_k = \Phi(\cdot)$ is mapped through the cell quantiles.

Proposition 1. *(a) For $\lambda = 1$ the marginal law of each member at each pixel is the cell's out-of-fold residual law; for $\lambda < 1$ the mixed score has unit variance but is not Gaussian (the standardised anomaly is a bounded sample statistic), so $\Phi(\cdot)$ of it is only approximately uniform and the marginal is reproduced only approximately — we measure the deviation (PIT audit, Section 5) and provide an exact variant that maps the score through the empirical CDF of the mixed score estimated on training folds; (b) for $\lambda > 0$ the members are exchangeable but not independent (the standardisation ties them: $\sum_k \tilde{a}_k = 0$), so fair scores, which assume independent members, are optimistic — we report the two-bank cross score of Roberts [2026] type as well; (c) if dates are exchangeable within cells, the out-of-fold quantile bands have finite-sample marginal coverage (split-conformal) and the ensemble is auto-calibrated with respect to the cell σ-algebra, being the unique minimiser of the expected proper score among laws measurable with respect to the cells; (d) its sharpness is bounded by $\mathbb{E}\,\mathrm{Var}(y \mid \textit{cell}) \geq \mathbb{E}\,\mathrm{Var}(y \mid x)$.*

Item (b) is not decorative: the variant $\lambda = 0$ (pure generator copula with pixelwise standardisation) achieved the best fair CRPS of all variants (40/40 dates) while its members were symmetric about

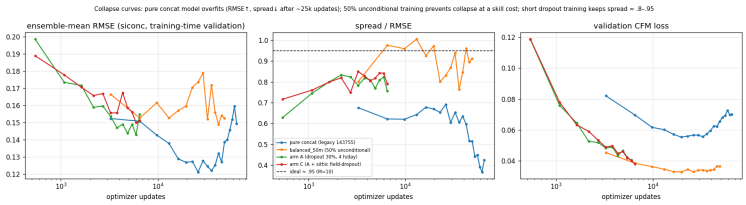


Figure 3: Training curves of the pure CFM model (validation loss, out-of-sample RMSE of the mean, spread/RMSE) and of the model with 50% unconditional steps.

the mean with no sampling variability, which the rank histogram exposed as U-shaped; fair scores assume exchangeability. The relation to EMOS, ensemble-copula coupling and the Gaussian-copula approach [Gneiting et al., 2005, Schefzik et al., 2013, Möller et al., 2013] is direct; what is new is the reason it works for a generative ensemble (Theorem 3) and the out-of-fold construction of the marginals on the evaluation dates themselves.

5 A real case: sea-ice data assimilation

Problem. The analysis of sea-ice concentration (and thickness) on day d over an Arctic subdomain (320×256 pixels) is generated from a background—the model field of the same date one year earlier—and observations on real Sentinel-3 SRAL tracks of days $d, d-1, d-2$ (OSSE: values are model truth on the real geometry; ≈ 700 pixels per day in ≈ 150 dotted segments, southern part of the domain only). The generator is a 50M-parameter UNet concat-conditioned CFM trained on 2016–2021 (six years), evaluated on 40 dates of 2022 (development) with 10 members; 2023 is sealed for a single final run. Two protocols exist: native (thickness visible) and "siconc-only" (thickness hidden, for comparison with a 3D-Var system).

No neighbours. The dominant uncertainty given the background is the interannual anomaly (73% of the anomaly variance at 20×16 scale is explained by year–month), of which the training set contains six realisations. Measured directly: for the 40 development dates, the distance from the background to the nearest training background (coarse scale) is 0.47 of the RMS anomaly to be predicted (median; 0.3–1.07), and the nearest neighbour is frequently from another month. This is the collapse regime of Fig. 1.

Collapse and constant spread. The pure CFM model's validation loss rises after epoch 5, its out-of-sample RMSE after epoch 7, and its spread decreases monotonically (spread/RMSE $0.68 \to 0.43$, Fig. 3); under the strict protocol the last checkpoint has spread/RMSE 0.49 and 80% coverage 0.31. A model trained with 50% unconditional steps does not collapse but its spread is a constant of the training set (0.135–0.141 in all four cells of year $\times$ protocol) while its error varies (0.156–0.190): spread/RMSE 0.90 on 2022 native, 0.76 on the training year 2019, 0.72 under the siconc-only protocol. A regularised retraining (condition dropout, early stopping by out-of-fold CRPS) reaches spread/RMSE 0.98 at the price of +6% RMSE. The in-sample test refutes memorisation for the balanced model (in-sample RMSE is not lower than out-of-sample).

Recalibrated ensemble (out of sample). Table 1 and Fig. 4 summarise the layer on the balanced generator, cross-fitted on the 40 development dates. Fair CRPS $0.0415 \to 0.0340$ (-18%, 38/40 dates, CI $[-0.0108, -0.0050]$), RMSE $0.156 \to 0.137$, spread/RMSE 0.99 in every distance

	background	background+layer	generator	generator+layer (v15)
RMSE of the mean	0.256	0.180	0.156	**0.137**
fair CRPS	–	0.0502	0.0415	**0.0340**
spread/RMSE	–	0.93	0.90	**0.99**
rank-histogram TV (per date)	–	0.077	0.157	0.085
RMSE >40 px from tracks	0.290	0.244	0.203	**0.197**

Table 1: Development year 2022, native protocol, 40 dates, 10 members, cross-fitted layer. The layer alone (on the background) is calibrated but the generator is needed as a conditional-mean and structure model (-32% CRPS).

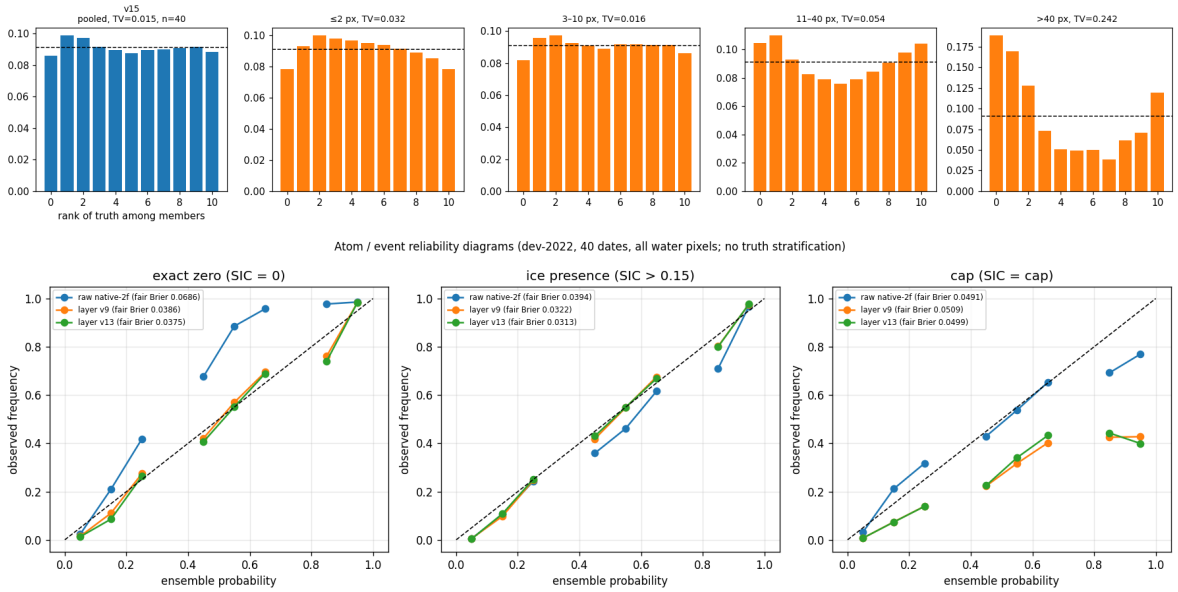


Figure 4: Rank histograms (pooled and by distance to the day-d track) of the recalibrated ensemble, and reliability of the atoms (exact zero, ice presence, cap) for generator vs generator+layer.

stratum, pooled rank-histogram TV 0.015 against a null tolerance of 0.034 for a perfect 10-member ensemble, 80% coverage 0.81–0.91, exact-zero atom Brier $0.069 \to 0.038$, day-d observations reproduced exactly. Multivariate scores agree: patch energy score -16% (35/40), variogram scores at lags 8–32 px improved, block-mean CRPS at 4/16/64 px $-21/-26/-34\%$, high-wavenumber power of the members $1.14\times$ the truth (generator 1.11). Under the siconc-only protocol CRPS improves by 31% (40/40). On the training year 2019 (a stress test) mean CRPS improves by 32%, but 14 winter dates are worse than the raw generator: the generator's conditional bias is seasonal (near zero in winter, -0.1 to -0.5 in summer) and the pooled cells transfer a summer shift to winter; a season axis fixes the mean (RMSE improved on 31/40 dates) but thins the cells on a single year.

Where the gain comes from (ablations at identical marginals). Table 2 decomposes the layer. Shifting the generator's members to the kriged mean and pinning the day-d observations, with no recalibration of the spread, already yields CRPS 0.0352 (76% of the total gain) and keeps the generator's fine-scale texture; the out-of-fold marginals add -3% CRPS and, mainly, calibration (per-date rank TV $0.13 \to 0.08$, spread/RMSE $\to 1.0$). With the marginals fixed, the generator's own-date dependence beats a stationary field and the dependence of another date in single-site

variant	fair CRPS (dates < raw)	ES 16×16	VS lag 1 / 16	CRPS block 1
raw generator	0.0415	0.821	0.0060 / 0.0271	0.0361
mean correction only (kriging + obs)	0.0352 (37)	**0.703**	**0.0058** / 0.0231	0.0270
+ marginals, stationary copula (λ=1)	0.0352 (34)	0.728	0.0066 / 0.0254	0.0277
+ marginals, own-date generator copula (λ=0)	**0.0340** (40)	0.725	0.0069 / 0.0243	0.0270
+ marginals, other-date generator copula	0.0351 (32)	0.723	0.0078 / 0.0244	0.0273
+ marginals, mixture (λ=0.5, v15)	0.0341 (38)	**0.689**	0.0067 / 0.0245	0.0271

Table 2: Ablations on the 40 development dates (native protocol, cross-fitted). All "+ marginals" rows share the same kriged mean and out-of-fold cell marginals and differ only in the copula. Differences in single-site CRPS between them measure the non-uniformity of the mixed score's PIT (Section 4).

CRPS (0.0340 vs 0.0352/0.0351), but only the mixture improves the energy score (0.689 vs 0.728 stationary and 0.725 own-date pattern); the pure generator pattern is over-dispersed (1.16) and rough. The honest summary is: the generator supplies a conditional mean and structure, a classical assimilation update supplies most of the skill, and the out-of-fold marginals supply calibration.

Standard post-processing baselines. Classical recalibration around the *raw* ensemble mean, fitted on the same training folds, changes little: EMOS-type variance inflation per distance stratum gives CRPS 0.0413, ensemble-copula coupling with the same out-of-fold cell marginals and the generator's member ranks 0.0408 (it does repair the zero atom, 0.41 → 0.53, and the 80% coverage, 0.76 → 0.88), a split-conformal band 0.0564 (no atoms, symmetric). The layer reaches 0.0335 (−19% against the best baseline, 39/40 dates) because it also corrects the mean by the observation innovations: the generator's defect is not a wrong spread scale but a conditional bias and a wrong support, which spread inflation cannot address.

Sampling-law audit. The probability-integral transform of the mixed score on held-out dates has KS distance to uniformity 0.01–0.02 for $\lambda = 1$, 0.05–0.06 for $\lambda = 0.5$ and 0.11–0.15 for $\lambda = 0$: the tails of the cell marginals are compressed unless the score is mapped through its own CDF, which we estimate on the training folds (final version). The two-bank cross score—the layer applied independently to two halves of the raw ensemble, scored across banks so that no independence within a bank is assumed—is 0.0329 against 0.0413 for the raw generator (better on 40/40 dates), so the reported gain does not rest on the exchangeability assumption of fair scores; the stratified-ensemble unit test of Roberts [2026] reproduces $M/[6(M+1)]$.

Is generativity needed? A deterministic UNet trained with the MSE objective on the same channels, architecture and budget (arm R; state channel zero, $t = 0$, 12 epochs) overfits from epoch 3 and reaches RMSE 0.227 on the 40 development dates (native protocol; 0.208 with the optimistic checkpoint selected on the same year), against 0.166 for the ensemble mean of the CFM model of the same series at its fixed final epoch (arm A) and 0.256 for the background. The generator is therefore needed as a *regressor*, not for its spread (which the layer rebuilds anyway): with one realisation per condition the MSE minimiser is also a point mass at the training truth (Theorem **??**), and the (y_0, t)-augmentation of the CFM objective acts as a regulariser that generalises better to new conditions.

Removing the identifying part of the condition. We retrained the same architecture with the background removed from the condition (tracks of days $d, d-1, d-2$ only, observation noise $\sigma = 0.05$ resampled at every step, 24 epochs; arm D), and a variant with real satellite geometry for every training sample and a realistic, spatially coherent, SIC-dependent error model fitted to SRAL–AMSR2 discrepancies (arm E). Without the key the spread no longer contracts with training (arm D: spread 0.164 → 0.143 while the RMSE falls 0.257 → 0.191 on 14 dates; spread/RMSE 0.64 → 0.75), in contrast to the pure model with the background (0.68 → 0.43). Under the strict protocol (40 dates, 10 members, adaptive solver) arm D reaches RMSE 0.171 against 0.156 for the same architecture with the background (0.256 for the background itself), spread/RMSE 0.86 and 80% coverage 0.79 over the domain, and is *calibrated near the tracks* (spread/RMSE 1.00, coverage 0.82, RMSE 0.098 vs 0.106 with the background). Removing the identifying part of the condition costs 10% RMSE, not the climatological skill one might fear; the residual under-dispersion sits where the condition carries no information and the model still reproduces particular training fields—the real track footprint of a training date is itself an exact, unique part of the condition (Corollary 1(e)). Randomising it (arm F: real curved footprints of random training-year dates for *every* training sample, evaluated on the real 2022 geometry) did *not* produce the calibration-by-construction that the corollary would predict for a nuisance identifier: RMSE 0.194 against 0.171 for arm D, spread/RMSE 0.60 against 0.90, worse in every month. The footprint is not information-free: through the 27-day repeat cycle of the orbit it is correlated with the season, so removing it costs information (the B term of the information-cost bound rather than the nuisance-averaging identity); a direct check—arm D evaluated on library geometry instead of its real geometry—is reported in the appendix. Arm E, trained on sparser realistic geometry, is worse overall (0.202 vs 0.191 on the 14-date curve) but better next to the tracks (0.100 vs 0.120), and did not converge in 24 epochs.

Sealed test year (2023, 200 consecutive days, 3D-Var protocol). Two runs exist on the same 200 days (2023-01-01 to 07-19, siconc-only, 5 members, background RMSE 0.275). The legacy generator gives analysis RMSE 0.157 and IIEE 0.118 → 0.082. The clean retrain used for the probabilistic line (arm C, field-dropout, checkpoint chosen on dev-2022) gives RMSE 0.203 (dev-2022: 0.256 → 0.181, i.e. the same −26 to −29% relative gain), spread 0.157, spread/RMSE 0.87, coverage-80 0.74, but IIEE 0.118 → 0.138: the ice-edge error is *worse* than the background in May–June (IIEE 0.12 → 0.18 and 0.17 → 0.26), while January–April behave as on dev. The spread of arm C is unchanged between years while its error grows—the generator does not know that 2023 is harder, which is exactly what the day-d track residuals (LOO nugget) in the layer are meant to correct. The probabilistic line is the single permitted contact with the test year: the frozen layer, fitted on the 40 dev dates of 2022 only, applied unchanged to the 200×5 saved arm-C ensembles (10 layer members). Fair CRPS 0.0575 → 0.0320 (−44%, paired-bootstrap CI $[-0.029, -0.022]$, 189/200 days), RMSE of the mean 0.203 → 0.145 (184/200; below the legacy deterministic analysis 0.157), spread/RMSE 0.77 → 0.98, per-date rank TV 0.130 → 0.067, zero mass 0.46 → 0.60 (truth 0.63), cap mass 0.041 → 0.068 (truth 0.027; the known over-censoring). By distance to the nearest track, spread/RMSE is 1.17 at 1–2 px and 1.05 beyond 40 px. By month, the gain is largest exactly where the generator was worst: May CRPS 0.074 → 0.036, June 0.100 → 0.040 (RMSE 0.294 → 0.172)—the day-d track residuals carry the information about 2023 that the generator lacks, and the kriging update plus OOF marginals propagate it over the domain. The colleagues' 3D-Var numbers are to be inserted by the authors.

6 Limitations

One out-of-sample year; the cap atom is not discriminated by any variant (observed frequency 0.44 at forecast probability > 0.9); the $> 40\,\mathrm{px}$ stratum has little statistics; the choice of cells is manual and the seasonal shift shows what a missing predictor costs; the copula class is a model, checked only by multivariate scores; the layer versions were selected on the same development dates (selection optimism of a few per cent CRPS, Cawley and Talbot, 2010); the test year 2023 was used once before for the deterministic comparison of the balanced generator, so the probabilistic run is the second contact with that year; the OSSE uses exact track values whereas real SRAL values have SIC-dependent, spatially coherent errors and are mostly unclassified in the marginal ice zone. Theorem 2 concerns the idealised minimiser; a real network adds its own implicit bandwidth. The smoother view for networks without explicit noise (2) is a hypothesis with two proven special cases.

7 Related work

Memorisation and generalisation in diffusion models. Kadkhodaie et al. [2023], Yoon et al. [2023], Somepalli et al. [2023] document the transition from memorising training images to generalising as the training set grows, along the data axis; our Theorems 1–2 are the same dichotomy along the *condition* axis, where its consequence is not reproduction of training images but a spurious posterior, and where a condition that identifies its target keeps the model in the memorising regime however large the dataset. Tang et al. [2024] give minimax rates for conditional diffusion estimators under smoothness; our setting is the one in which the smoothness is not resolved by the spacing of the conditions. *Conditioning augmentation.* Noise on the conditioning signal is used in cascaded and super-resolution diffusion [Ho et al., 2022] as a heuristic; Theorem 2 states what it estimates and when it helps. *Ensemble post-processing.* EMOS [Gneiting et al., 2005], ensemble-copula coupling [Schefzik et al., 2013], the Schaake shuffle and the Gaussian-copula approach [Möller et al., 2013] supply the building blocks of Section 4; fair scores and their pitfalls with non-independent members are due to Ferro [2014], Roberts [2026], and selection bias in model selection to Cawley and Talbot [2010]. *Generative weather and assimilation.* GenCast [Price et al., 2025] is the calibrated case of the smoother view; CorrDiff-type regression-plus-residual-diffusion designs are the training-side analogue of our two-stage decomposition and inherit its two requirements (out-of-fold residuals, non-identifying residual conditions).

References

Y. Lipman, R. T. Q. Chen, H. Ben-Hamu, M. Nickel, M. Le. Flow matching for generative modeling. ICLR 2023.

A. Tong et al. Improving and generalizing flow-based generative models with minibatch optimal transport. TMLR 2023.

I. Price et al. Probabilistic weather forecasting with machine learning. Nature 2025.

A. Jacot, F. Gabriel, C. Hongler. Neural tangent kernel. NeurIPS 2018.

P. Deheuvels. La fonction de dépendance empirique et ses propriétés. 1979.

C. Sempi. Convergence of copulas: critical remarks. 2004.

R. Schefzik, T. L. Thorarinsdottir, T. Gneiting. Uncertainty quantification in complex simulation models using ensemble copula coupling. Statistical Science 2013.

T. Gneiting, A. E. Raftery, A. H. Westveld, T. Goldman. Calibrated probabilistic forecasting using ensemble MOS. MWR 2005.

A. Möller, A. Lenkoski, T. L. Thorarinsdottir. Multivariate probabilistic forecasting using ensemble Bayesian model averaging and copulas. QJRMS 2013.

Z. Kadkhodaie, F. Guth, E. P. Simoncelli, S. Mallat. Generalization in diffusion models arises from geometry-adaptive harmonic representations. 2023.

T. Yoon et al. Diffusion probabilistic models generalize when they fail to memorize. 2023.

G. Somepalli et al. Diffusion art or digital forgery? CVPR 2023.

R. Tang, Z. Lin, Y. Yang. Conditional diffusion models are minimax-optimal and manifold-adaptive for conditional distribution estimation. arXiv:2409.20124, 2024.

C. D. Roberts. Ensemble-size-dependence of deep-learning post-processing methods that minimize an (un)fair score. arXiv:2602.15830, 2026.

C. A. T. Ferro. Fair scores for ensemble forecasts. QJRMS 2014.

G. C. Cawley, N. L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. JMLR 2010.

J. Ho et al. Cascaded diffusion models for high fidelity image generation. JMLR 2022.